Stats & Stumps: A Novel Impact Factor for Player Comparison and Match Prediction in Cricket

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Abstract

Cricket's fastest-growing format, Twenty20 Internationals (T20Is), presents unique challenges for evaluating players and predicting match outcomes. Team selectors often justify decisions based on "experience", yet performance in other formats (e.g., Test cricket) does not reliably translate to T20s. This project builds quantitative, venue-adjusted impact metrics from batting and bowling statistics (runs, strike rate, boundaries, wickets, economy, maiden overs) to provide objective alternatives for player and team evaluation. Using a dataset of 1,061 T20Is, four approaches were tested: a heuristic, logistic regression with regularization, random forests, and support vector machines (SVMs). Logistic regression and SVMs achieved the best accuracy (~72%), remaining interpretable while balancing RMSE and Log-Loss. The final model was backtested on the 2024 T20 World Cup, correctly characterizing the close India–South Africa final. An interactive app enables player comparison, model backtesting of all historical matches, and match predictions for a user-generated game. Overall, the framework shows how statistical modeling can complement selection decisions and improve T20I outcome prediction.

Keywords:
T20I, Cricket, Machine Learning, Player Impact, Cricket Analytics, Venue Adjustment, Regression

Reproducibility Statement: The code used in this project is available in Github, at https://github.com/ArchithSharma/CricketPredictions. In addition, a published version of the R Markdown used to generate figures and conduct analysis is present at https://rpubs.com/ArchithS/CricketData. The current version of the code is in the **Markdown_Script** folder in the Github repository.

Web Application: https://archithsharma.shinyapps.io/Cricket-Analyzer

1 Introduction

Cricket is a bat-and-ball sport that originated in England and has grown into one of the most popular sports worldwide, particularly in countries like India, Australia, South Africa, and Pakistan. Played between two teams of 11 players each, the game involves batting, bowling, and fielding, with the primary objective being to score more runs than the opposing team. Cricket is played in different formats, ranging from the traditional five-day Test matches to the fast-paced Twenty20 (T20) format.

T20 cricket, the shortest and most explosive format of the game, revolutionized the sport with its fast-paced action and entertainment value. Each team gets a maximum of 20 overs (120 balls) to bat, encouraging aggressive stroke play, quick scoring, and strategic bowling. Since its official introduction in 2003, T20 cricket has gained massive popularity, leading to global tournaments such as the ICC T20 World Cup and franchise-based leagues such as the Indian Premier League (IPL) and the Big Bash League (BBL). With its thrilling finishes, power-hitting, and emphasis on adaptability, T20 cricket has attracted a new generation of fans while maintaining the essence of the sport.

The rise of predictive analytics in cricket has closely coincided with the rapid growth of the T20 format. With T20 matches fast-paced and often decided by fine margins, teams, analysts, and betting markets have increasingly turned to data-driven approaches to gain a competitive edge. The shorter format requires quick decision making, making real-time data analysis and predictive modeling crucial to optimize team strategies, player selection, and match predictions.

In this paper, a unique approach will be taken that focuses on a new impact factor (IF) of players for both batting and bowling, in addition to consideration of match venues. These factors will then be used to train 3 types of models commonly used in sports prediction: Regularized logistic regression, random forest classifier, and support vector machines (SVMs), with high test set accuracies of 70 - 72%. (Vistro et al., 2019)

The paper is organized as follows. Section 2 will explore the Cricinfo dataset and derive the impact factor used for batting and bowling, along with a ground-based scaling to account for different venues. Afterwards, in Section 3 the comprehensive impact factors are adjusted for model training. Multiple models are tested on the adjusted factors in Section 4. Section 5 shows the accuracy percentages of the models along with top players and venues according to the impact factor, followed by a discussion of the models in Section 6. A description of the developed web application is included in Section 7, followed by conclusions and a discussion of the model limitations in Section 8. Finally, there is an appendix containing a definition of common cricket terms used throughout the paper and the list of test nations (Appendix).

2 Data and Impact Factor

All data pre-processing, analysis and modeling was performed in R. (R Core Team, 2024) See the Reproducibility Statement to locate the code used on GitHub and regenerate the figures.

2.1 Description of Data

	Date	Player	Country	Runs	NotOut	Minutes	BallsFaced	Fours	Sixes	StrikeRate
1	2018-07-03	AJ Finch	AUS	172	FALSE	NA	76	16	10	226.3158
2	2019-02-23	Hazratullah Zazai	AFG	162	TRUE	NA	62	11	16	261.2903
3	2013-08-29	AJ Finch	AUS	156	FALSE	70	63	11	14	247.6190
4	2016-09-06	GJ Maxwell	AUS	145	TRUE	NA	65	14	9	223.0769
5	2024-01-17	FH Allen	NZL	137	FALSE	94	62	5	16	220.9677
6	2025-02-02	Abhishek Sharma	IND	135	FALSE	93	54	7	13	250.0000
7	2023-02-01	Shubman Gill	IND	126	TRUE	99	63	12	7	200.0000
8	2017-07-09	E Lewis	WI	125	TRUE	NA	62	6	12	201.6129
9	2016-01-31	SR Watson	AUS	124	TRUE	86	71	10	6	174.6479
10	2023-11-28	RD Gaikwad	IND	123	TRUE	97	57	13	7	215.7895
11	2012-09-21	BB McCullum	NZL	123	FALSE	72	58	11	7	212.0690
12	2022-09-08	V Kohli	IND	122	TRUE	90	61	12	6	200.0000
13	2021-04-14	Babar Azam	PAK	122	FALSE	77	59	15	4	206.7797
14	2024-01-17	RG Sharma	IND	121	TRUE	103	69	11	8	175.3623
15	2024-02-11	GJ Maxwell	AUS	120	TRUE	70	55	12	8	218.1818
16	2024-11-15	NT Tilak Varma	IND	120	TRUE	69	47	9	10	255.3191
17	2015-01-11	F du Plessis	RSA	119	FALSE	99	56	11	5	212.5000
18	2023-12-19	PD Salt	ENG	119	FALSE	80	57	7	10	208.7719
19	2016-01-10	Mohammad Shahzad	AFG	118	TRUE	NA	67	10	8	176.1194
20	2017-12-22	RG Sharma	IND	118	FALSE	61	43	12	10	274.4186
21	2023-03-26	J Charles	WI	118	FALSE	60	46	10	11	256.5217

Figure 1: Batting Data

The data is taken from ESPN Cricinfo, a reputable source containing in-depth statistics on all T20Is played and the players themselves. (ESPN, 2025a) To access the data in R, the `cricketdata` package made by Rob Hyndman (Hyndman et al., 2025) was used to access the Statsguru data in R as a dataframe. A screenshot of the batting dataframe is shown above in Figure 1.

The data is then filtered to only include T20Is between the test-playing full ICC members, defined in the appendix. The data is also joined by match to determine the total impact factor of teams in their matches, both bowling and batting.

2.2 Derivation of Impact Factor

At the core of this analysis is the novel impact factor used in comparing batting and bowling impacts. To determine the formula for batting and bowling impact, correlation to the target metric between the total impact of both teams was prioritized in addition to similarity in the distribution of the impact factors each game.

The impact factor was manually tuned to a quantity that sufficiently addressed both criteria. They are defined below in Equation 1 & Equation 2:

2.2.1 Definition of Bat and Bowl Impact

$$\text{Batting Impact} = (R \times 0.125) + ((SR - 130) \times 0.025) + (4s \times 0.3) + (6s \times 0.5) - \text{OutPenalty} \quad (1)$$

$$\begin{aligned} \text{where:} \quad & R = \text{Runs scored} \\ & SR = \text{Strike rate} = \left(\frac{\text{Runs}}{\text{Balls Faced}} \times 100 \right) \\ & 4s = \text{Number of fours hit} \\ & 6s = \text{Number of sixes hit} \\ & \text{OutPenalty} = \text{Fixed penalty of 0.5 if the batter is dismissed} \end{aligned}$$

Example:

A batter scores $R = 40$ runs off 28 balls, hits 3 fours, 2 sixes, and is dismissed.

$$SR = \frac{40}{28} \times 100 \approx 142.86, \quad \text{OutPenalty} = 0.5$$

$$\text{Batting Impact} = (40 \times 0.125) + ((142.86 - 130) \times 0.025) + (3 \times 0.3) + (2 \times 0.5) - 0.5$$

$$= 5 + (12.86 \times 0.025) + 0.9 + 1.0 - 2.5 = 5 + 0.3215 + 0.9 + 1.0 - 0.5 \approx \mathbf{6.72}$$

$$\text{Bowling Impact} = (W \times 3.25) + (M \times 3) - ((E - 7.4) \times 0.5) \quad (2)$$

$$\begin{aligned} \text{where:} \quad & W = \text{Number of wickets taken} \\ & M = \text{Number of maiden overs bowled} \\ & E = \text{Economy rate} = \left(\frac{\text{Runs Conceded}}{\text{Overs Bowled}} \right) \\ & 7.4 = \text{Baseline economy rate for adjustment} \\ & 0.5 = \text{Penalty per unit increase in economy above 7.4} \end{aligned}$$

Example:

A bowler takes $W = 2$ wickets and a $E = 6.85$ economy rate and has no maiden overs.

$$\text{Bowling Impact} = (2 \times 3.25) + (0 * 3) - ((6.85 - 7.4) * 0.5) = 6.5 + 0.55 * 0.5 \approx \mathbf{6.775}$$

Both batting and bowling equations consider a baseline adjustment for strike rate and economy rate based on the average score of around 149 in T20Is. (Statsguru, 2025) Both example players did very well in their respective games and had high similar impacts of around 6.75, showing the relevance of the heuristic.

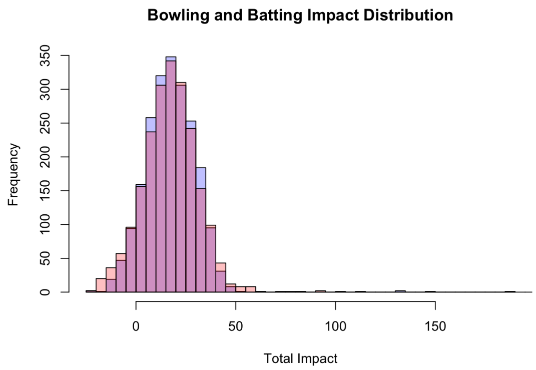
Afterwards, the batting and bowling impacts were grouped by their totals by game for each team. Afterwards, the differences between total impact factors are used as a heuristic to be correlated to the winner of the match (interpreted numerically as 1, 0). As can be seen in the code, the correlation value is 0.774 for 1059 T20Is, and the R^2 is 0.599, approximately 0.6 which indicates high explainability.

2.2.2 Validation of Bat/Bowl Similarity

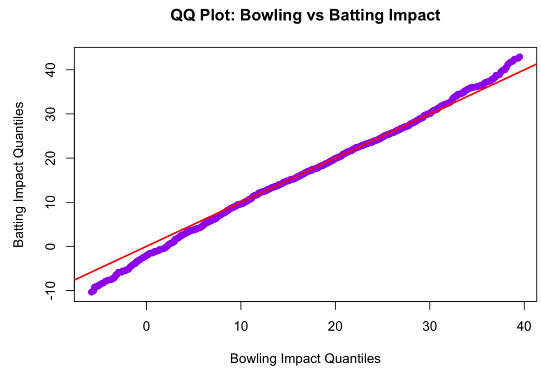
To validate the assumption that the distributions are similar, a Kolmogrov-Smirnov test is conducted and the Cohen's d value is measured. and the result is displayed in Table 1. The $p = 0.37$ and Cohen's $d = 0.03 < 0.2$ minimal effect size shows that the distributions are very similar. (Cohen, 1988) (Berger & Zhou, 2014) In addition, the visual validation of the histogram and QQ plot in Figure 2 shows how similar the total impact factors are, and that they can be used for modeling.

Statistic	Value	Notes
Kolmogorov–Smirnov D	0.0281	Two-sample, two-sided
p-value	0.3678	Not significant ($\alpha = 0.05$)
Cohen's d	0.0278	Negligible effect size

Table 1: Kolmogorov–Smirnov Test and Effect Size Results



(a) Impact Distribution Histogram



(b) Impact Distribution QQ-plot

Figure 2: Histogram and QQ-Plot of Batting/Bowling Impact

3 Determining Impact Factors to be used in Prediction

In this study, the performance of players in a T20 match is evaluated through a series of factors that encompass both their batting and bowling abilities. The impact of each player is adjusted for various game-specific conditions, including venue-based adjustments and player rankings. Below, the rationale behind the decision-making process is outlined for each factor used in the calculation of player impact.

3.1 Player Ratings and Impact Calculation

Each player's performance is assessed based on their batting and bowling ratings, which are derived from historical performance data. These ratings are extracted from the full player dataset, which includes batting impact (`BatImpactperGame`) and bowling impact (`BowlImpactperGame`) for each player. These ratings reflect the overall contribution of a player to their team's performance in previous matches.

3.2 Ground Buffs (Venue Factors)

The performance of players is also adjusted based on the specific characteristics of the ground where the match is played. Each ground has unique attributes that can influence player performance, such as pitch conditions, weather, and altitude. The `venue_factors` dataset provides ground-specific adjustments known as "ground buffs". These buffs are applied to both batting and bowling ratings to account for the venue's impact on player performance. The following ground buffs are considered:

- **Batting Buff**: This factor adjusts the batting impact based on ground-specific conditions such as pitch type, boundary size, and weather.
- **Bowling Buff**: This factor adjusts the bowling impact based on similar ground conditions that affect how bowlers perform, including pitch conditions and boundary dimensions.

3.3 Weighting by Player Ranking

The weight assigned to each player's performance is influenced by their relative ranking in the team. Batting and bowling impact are adjusted according to player rankings:

- **Batting Impact**: The top 8 batsmen (top and middle order) in the playing XI receive a weighted adjustment to their batting impact. The adjustment is proportional to the player's ranking, with higher-ranked players receiving a greater weighting. The weight is calculated as a function of the player's position in the batting order, with later batsmen receiving lower weights to reflect their reduced role in the match.

- **Bowling Impact**: Similarly, the top 6 bowlers are given weighted adjustments to their bowling impact, based on their ranking. The adjustment is calculated in a similar manner, with higher-ranked bowlers receiving greater weight. The bottom bowlers are excluded from the impact calculation, as they are less likely to bowl in the match.

3.4 Impact Per Game Adjustments to be used for prediction

Once the initial ratings are adjusted for venue buffs and player rankings, the resulting impact values, batting and bowling are calculated for each player. The calculations are performed iteratively for both teams, with batting and bowling impacts adjusted according to the above factors:

- For batsmen, the weight is scaled according to their position in the batting order.
- For bowlers, a similar scaling is applied, with the most prominent bowlers receiving higher weights.
- The adjusted values for each player are stored and used to update the overall team impact.

3.5 Calculation of Total Player Impact

Finally, the adjusted batting and bowling impacts are compiled into a total impact score for each player. This score is split into batting and bowling impacts, with the first 8 players in the batting order contributing to the batting total, and the top 6 bowlers by impact score contributing to the bowling total. The impact scores for each player from both teams are then consolidated into a total impact matrix, which represents the overall performance potential of the two teams based on individual player ratings and match conditions.

In summary, the impact factors in this methodology are derived from a combination of player rankings, ground-specific adjustments, and positional weights. These factors provide a comprehensive assessment of player performance, accounting for both individual ability and contextual factors, such as venue and team composition.

4 Modeling

To predict T20I match outcomes based on player-level performance metrics, several classification models were implemented and evaluated using a dataset where each match was represented by the total impact scores of the top 8 batsmen and 6 bowlers per team. Even though there are only 11 players, there is overlap between these categories (all-rounders). There are 28 data cells, 8 batsmen and bowlers per team. As a baseline, a simple heuristic predicted the team with the higher cumulative impact score as the winner:

$$\hat{y} = \begin{cases} 1 & \text{if } \sum_{i=1}^{14} \text{Impact}_{i,\ \text{Team1}} > \sum_{j=1}^{14} \text{Impact}_{j,\ \text{Team2}} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

This naïve method yielded an accuracy of **65.98%**, providing a benchmark for more advanced models.

A logistic regression model with elastic net regularization was trained using 5-fold cross-validation to tune the hyperparameters α and λ. (Tay et al., 2023) The logistic model estimates the probability of Team 1 winning as:

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}))} \tag{4}$$

with coefficients obtained by minimizing the penalized negative log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ -\frac{1}{n} \sum_{i=1}^{n} \Big[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \Big] + \lambda \Big(\alpha \|\boldsymbol{\beta}\|_1 + (1 - \alpha) \tfrac{1}{2} \|\boldsymbol{\beta}\|_2^2 \Big) \right\} \tag{5}$$

Repeating this process over 30 iterations, the logistic regression model achieved a maximum accuracy of **71.7%**, and the lowest composite score of **1.318**, defined as:

$$\text{Composite Score} = (1 - \text{Accuracy}) + \text{RMSE} + \text{Log-Loss} \tag{6}$$

A random forest model with 500 trees was also evaluated, tuning the `mtry` parameter via 5-fold CV. (Kuhn, 2008) Random forest predictions are the majority vote of an ensemble of B trees:

$$\hat{y} = \text{mode}\left\{ T_b(\mathbf{x}) \,:\, b = 1, \ldots, B \right\} \tag{7}$$

The best random forest model achieved an accuracy of **70.28%** and a composite score of **1.364**.

Lastly, a linear Support Vector Machine (SVM) was evaluated using a grid of cost values $C = 2^{-5}, \ldots, 2^5$ and 5-fold cross-validation. (Meyer et al., 2023) The linear SVM learns a decision boundary:

$$f(\mathbf{x}) = \text{sign}\left(\mathbf{w}^\top \mathbf{x} + b \right) \tag{8}$$

with parameters $(\mathbf{w}, b)$ estimated by:

$$\min_{\mathbf{w}, \xi} \; \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^{n} \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \;\; \xi_i \geq 0 \tag{9}$$

The best SVM model reached an accuracy of **70.75%** with a composite score of **1.323**, comparable to the random forest.

Overall, logistic regression with elastic net regularization offered the strongest performance under the composite score metric, while also allowing for interpretability through coefficient inspection. This suggests that linearly weighted combinations of individual player impacts are highly predictive of T20I match outcomes.

5 Results

5.1 Player and Venue visualizations

Now that the impact factors are clearly defined, they can be used to construct rankings of players by batting, bowling, or all-around play and venue factors. Similarly, teams can be ranked by their recent impact factors. The full visualizations of the Top 15 in Career Impact and Impact per Game for batsmen are found in the images file in the GitHub repository, and can be generated from the R markdown.

All-rounders are defined as individuals in the top two-thirds of all players in batting and in the top third percentile of all players in bowling. The average venue impact was also min-max scaled to be used as a multiplier in the predictive modeling phase. Note the inclusion of players on multiple lists, such as Rashid Khan, Shakib al Hasan, and Wanindu Hasaranga being top all-rounders due to their bowling, and Virat Kohli consistently accumulating impact at a rapid pace over his career with the highest career impact and 3^{rd} highest per game batting impact.

Table 2 and Table 3 include the Top 5 in Batting Impact, Bowling Impact over the span of a career, in addition to the Top 5 in those categories per game. The total impact of all-rounders is also calculated and displayed in Table 4. Lastly, the top 5 venues for Bowling and Batting are in Table 5.

Rank	Player	Career Impact	Player	Impact per Game
1	V Kohli (IND)	608.06	SA Yadav (IND)	5.46
2	RG Sharma (IND)	599.65	NT Tilak Varma (IND)	5.39
3	JC Buttler (ENG)	572.2	V Kohli (IND)	5.11
4	Babar Azam (PAK)	530.37	KP Pietersen (ENG)	5.08
5	MJ Guptill (NZL)	511.31	YBK Jaiswal (IND)	5.03

Table 2: Batting Impact Top 5: Career & Per Game

Rank	Player	Career Impact	Player	Impact per Game
1	Rashid Khan (AFG)	472.77	Rashid Khan (AFG)	6.06
2	TG Southee (NZL)	448.67	Kuldeep Yadav (IND)	5.96
3	Shakib Al Hasan (BAN)	421.38	BAW Mendis (SL)	5.95
4	AU Rashid (ENG)	412.93	Imran Tahir (RSA)	5.74
5	IS Sodhi (NZL)	400.5	PW Hasaranga (SL)	5.2

Table 3: Bowling Impact Top 5: Career & Per Game

Rank	Player	Total Impact	Player	Impact per Game
1	Shakib Al Hasan (BAN)	658.5	Rashid Khan (AFG)	6.57
2	GJ Maxwell (AUS)	548.05	SR Watson (AUS)	6.23
3	HH Pandya (IND)	544.81	Shakib Al Hasan (BAN)	5.88
4	Shahid Afridi (PAK)	527.63	PW Hasaranga (SL)	5.88
5	Rashid Khan (AFG)	512.78	Shahid Afridi (PAK)	5.61

Table 4: All-Rounder Impact Top 5: Career & Per Game

Rank	Top Bowling Grounds	Top Batting Grounds
1	Dallas	Guwahati
2	New York	Hyderabad
3	Kingstown	Bloemfontein
4	Colombo (PSS)	Brabourne
5	Port of Spain	Mohali

Table 5: Top 5 Batting/Bowling Venues

5.2 Model Results

The modeling results are organized below in Table 6.

Model	Accuracy	RMSE	Log-Loss	Composite Score	95% Confidence Interval
Heuristic	0.6598	NA	NA	NA	NA
Logistic Regression	0.7170	0.4474	0.5876	1.3181	(1.2997, 1.3364)
Random Forest	0.6840	0.4519	0.5959	1.3639	(1.3391, 1.3887)
SVM	0.7028	0.4441	0.5813	1.3226	(1.3017, 1.3434)

Note: Composite Score is calculated as (1 - Accuracy) + RMSE + Log-Loss. Lower values indicate better overall model performance.

Table 6: Model performance based on mean accuracy and 95% confidence intervals over 30 iterations

6 Discussion

6.1 Analysis of Players and Venues

6.1.1 Player Analysis

Virat Kohli, often called the world's best batsman, ranks third all-time in runs and second in centuries (ESPN, 2025b) Critics question his T20I strike rate (TOI Sports Desk, 2024), yet the impact factor ranks him the top batsman overall and third on a per-game basis. His ability to accumulate runs offsets strike rate concerns, showing the metric captures his role in anchoring innings.

6.1.2 Player Analysis: Ajantha Mendis

Ajantha Mendis (labeled BAW Mendis), Sri Lanka's "mystery spinner" (2008–2014), twice took six wickets in a T20I and was named Emerging Player of 2008. Once figured out, he struggled with consistency and lost his place to the accomplished Rangana Herath and pacers such as Nuwan Kulasekara (Cricbuzz, 2025).

The impact factor overestimates him, ranking him as high as third, since it lacks time weighting and inflates small but extreme records. Incorporating a random effect proportional to matches played could reflect inconsistency: Mendis' estimate, with one-third of Kohli's matches, would have about three times the variance. Because inconsistent players are dropped, the model remains representative.

6.1.3 Player Analysis: Glenn Maxwell

Glenn Maxwell exemplifies T20 versatility, combining power hitting with his 'golden-arm' off-spin. Most of his impact comes from batting, where unorthodox stroke play drives high scores. (ESPN, 2023), and his high batting impact scores and semi-decent bowling performances push him up the all-rounder list.

Compared to Rashid Khan, mainly a bowler with batting ability down the order, and Hardik Pandya, balanced but middling in both, the method assigns all three distinct yet high all-rounder impacts.

6.1.4 Venue Analysis: England vs. Subcontinent Venues

Venue analysis aligns with common views: 12 of the top 20 batting venues are in India or Pakistan, reinforcing their reputation as batting-friendly. English grounds show more balance; Lord's ranks 80th of 105 for batting, aided by its famous slope, while The Oval, Southampton, and Headingley fall in the 60–70 range. These results confirm that venue scaling from impact factors reflects real ground conditions.

6.1.5 Team Rankings Based on Recent Matches

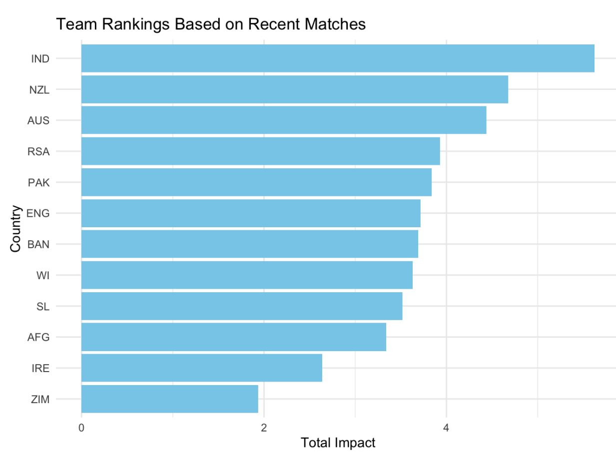


Figure 3: Rankings based on performance for last 10 matches

Based on the impact factor, rankings based on the total accumulated impact can be constructed for a team's last 10 matches in Figure 3. The data shows that India is the most in-form team, followed by New Zealand and Australia. South Africa are close behind, with Ireland and Zimbabwe well behind the rest. In the context of match records, the impact factor method does well in assessing a team's performance.

6.2 Analysis of Models

Four approaches were tested for predicting T20I outcomes: a heuristic model, logistic regression, random forests, and support vector machines (SVM). The heuristic, which picked the team with the higher total impact, reached 66% accuracy as a baseline. Logistic regression performed best, with 71.7% accuracy and the lowest composite score (1.318, 95% CI: 1.300–1.336). Random forests and SVMs followed at 68.4% (1.364, 95% CI: 1.339–1.389) and 70.28% (1.323, 95% CI: 1.302–1.343), respectively (Table 6).

Logistic regression was chosen for final prediction due to its strong composite score and interpretability. The model outputs probabilities, enabling uncertainty quantification—for example, it gave India a 53% win chance in the 2024 T20 World Cup Final against South Africa, aligning with the close 7-run result (Figure 5). Overall, logistic regression combines competitive accuracy with actionable probabilities, while its coefficients provide insight into how player features shape team-level predictions.

7 Application

I developed an application to compare players and show how the model can make predictions on past and future matches. With all the player database used to train the model, one part of the application features a comparison tool for players, where two player can be compared based on their impact and impact for game, as shown in Figure 4.

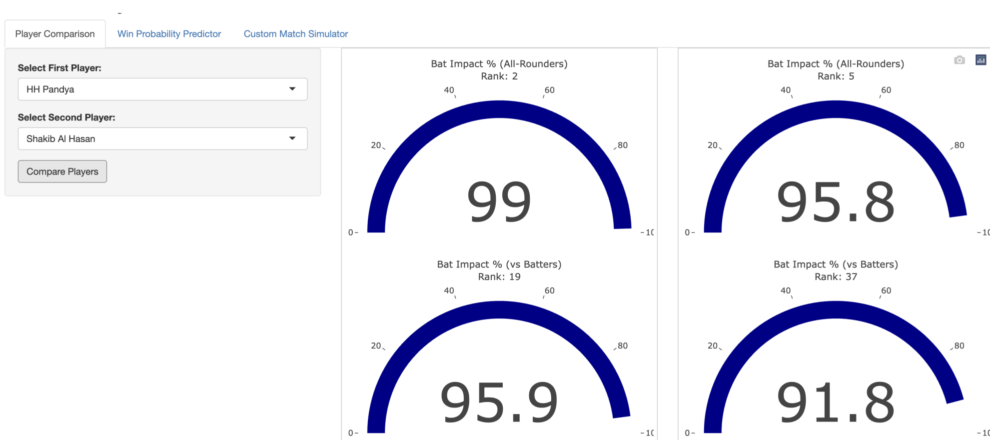


Figure 4: Comparison between Hardik Pandya & Shakib al Hasan in Web App

Another tab of the application measures the model's output on past matches. As mentioned before, the model's output in the app for the 2024 T20WC final is in Figure 5, and can be tested against all games in the database.

Lastly, to predict future matches or custom matches, the last tab of the app will take in 2 playing XIs and a venue, and output a win probability. An example was tested for the most recent T20I match between Afghanistan and Pakistan. The model's most recent data update was on August 16, 2025, and the match was played on August 29, 2025. The model predicted a 59.3% win probability for Pakistan, who went on to win the match by 39 runs, further legitimizing the model as shown in Figure 6.

The app is available at:

https://archithsharma.shinyapps.io/Cricket-Analyzer

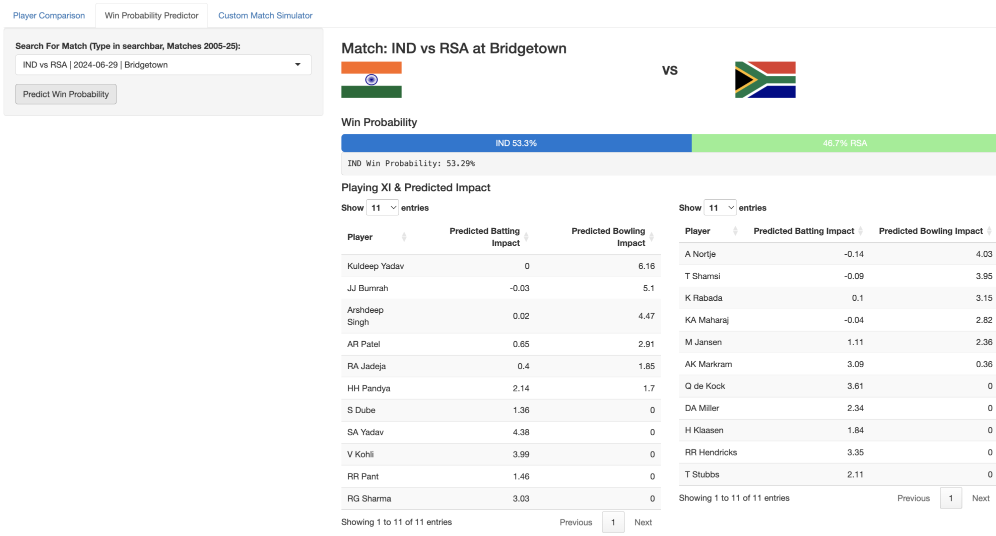


Figure 5: Past Match Model Output
IND vs RSA, T20 WC Final 2024

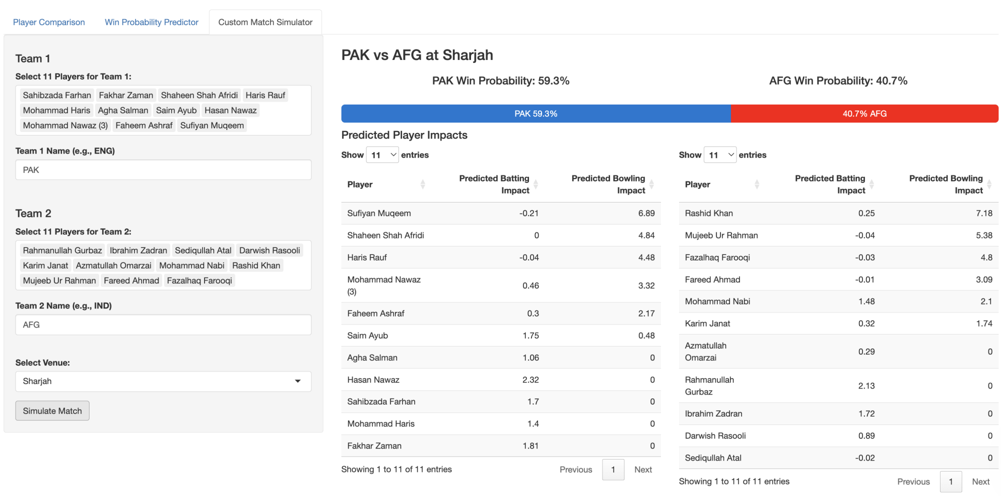


Figure 6: Predict Future Match based on Playing XI
AFG vs PAK, T20I, August 29 2025

8 Conclusions and Future Work

This paper has been maximizing the value of a simple model in T20I prediction. More complex models can consider strategy with the toss before matches, as many pitches get better to bat on because of the dew in the night, meaning there is a significant advantage in batting second. (Krishnaswamy, 2022) Recent player form can change the parameter passed into the logistic regression. For example, Virat Kohli was in a bad run of form during the T20 World Cup, only crossing 10 runs twice in the 7 matches before the final. Specific bowler-batsman combinations are not considered either, such as the Indian top order's issues with left arm pace bowlers. (Iyer, 2023) In addition, win probability could be mapped to a margin of victory as determined by the impact factor difference. Overall, however, this paper's methods have had high accuracy in predictions and an explainable impact factor that can be used to quantify team performance. This framework can streamline cricket analysis and objectively pit players and teams against each other through the power of data.

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Appendix: Glossary of Cricket Terms Used

For those familiar with baseball: Key parallels and differences explained

Batting Terms

Runs Total number of runs scored by a batsman.
Baseball parallel: Similar to total bases or RBIs, but batsmen accumulate runs throughout their time at bat rather than per at-bat.

Strike Rate The average number of runs scored per 100 balls faced. A higher strike rate indicates more aggressive scoring.
Baseball parallel: Like batting average but measuring offensive output per pitch faced. A 150 strike rate means 1.5 runs per ball, showing aggressive hitting.

Boundary A scoring shot that results in four (4) or six (6) runs.
Baseball parallel: Similar to extra-base hits, but automatic scoring based on where the ball lands.

Fours & Sixes A four occurs when the ball crosses the boundary after touching the ground; a six is when it crosses without touching the ground.
Baseball parallel: Fours are like ground-rule doubles that automatically score 4 runs; sixes are like home runs that score 6 runs.

Batting Impact Score A custom metric combining runs, strike rate, and boundaries to evaluate batting effectiveness.
Baseball parallel: Like OPS (On-base Plus Slugging) or WAR - a comprehensive offensive metric.

Top Order The group of batsmen who come out to bat first and score most of the runs, typically the first three or four players.
Baseball parallel: Similar to the top of the batting order (1-4 hitters), but these players may bat for much longer periods.

Middle Order The group of batsmen that are typically all-rounders and power hitters to try and get quick runs near the end.
Baseball parallel: Like cleanup hitters and power bats (5-7 in the order), focused on driving in runs and hitting for power.

Bowling Terms

Wickets The number of batsmen a bowler dismisses.
Baseball parallel: Like strikeouts or outs recorded by a pitcher, but more valuable since each dismissal is significant.

Maiden Overs Overs in which no runs are conceded.
Baseball parallel: Like a perfect inning where no runs score, but refers to 6 consecutive pitches (1 over) rather than 3 outs.

Economy Rate Runs conceded per over bowled. A lower economy indicates more efficient bowling.
Baseball parallel: Similar to ERA (Earned Run Average), measuring runs allowed per unit of pitching.

Spinner A bowler who bowls slow but gets the ball to turn, deceiving batsmen.
Baseball parallel: Like a knuckleball pitcher or changeup specialist - relies on movement and deception rather than velocity.

Pacer A bowler who bowls quick to try and beat batsmen with variations and leave them helpless.
Baseball parallel: Like a power pitcher (fastball, slider) who relies on velocity and hard breaking balls.

Bowling Attack The group of bowlers in a team and their combined effectiveness.
Baseball parallel: Similar to a pitching staff or rotation, but all bowlers may be used in a single game.

Bowling Impact Score A weighted metric assessing bowling performance using wickets, maidens, and economy rate.
Baseball parallel: Like a pitcher's Game Score or advanced metrics that combine multiple pitching statistics.

Match Context

T20I Twenty20 International – a short-format cricket match where each team bowls a maximum of 20 overs.
Baseball parallel: Each team gets 20 "innings" of 6 pitches each (120 total pitches), roughly equivalent to a 7-9 inning baseball game in time.

Venue Factor A numerical measure of how batting-friendly or bowling-friendly a stadium is, based on its average scoring.
Baseball parallel: Similar to park factors that measure whether a ballpark favors hitters or pitchers.

Playing XI The eleven players selected to participate in a match.
Baseball parallel: Like the starting lineup, but includes both position players and pitchers since cricket players both bat and field.

Strategy Terms

All-Rounder A player who contributes significantly with both bat and ball.
Baseball parallel: Like a two-way player (pitcher who can hit) such as Shohei Ohtani, but more common in cricket.

Toss The coin toss where the captain who wins decides whether to bat or bowl first.
Baseball parallel: Similar to home field advantage choice, but more impactful since field conditions heavily influence the decision.

Run Rate Runs per over (6 balls).
Baseball parallel: Like runs per inning, measuring the pace of scoring throughout the game.

Pitch Surface that the bowlers bowl to the batsmen on, can heavily influence team decisions by favoring batting or a certain kind of bowling.
Baseball parallel: Like field conditions, but much more influential - equivalent to if the mound height or distance changed significantly between games - cricket bowls are allowed to bounce on the surface and move around a lot more than baseball pitches.

Test Playing Nations

The 12 countries that compete at the highest level of international cricket: Afghanistan, Australia, Bangladesh, England, India, Ireland, New Zealand, Pakistan, South Africa, Sri Lanka, West Indies, Zimbabwe.

Key Differences for Baseball Fans

- **No foul territory**: Balls can be hit in any direction (360°)
- **No automatic outs**: Batsmen stay in until dismissed
- **Continuous play**: Batsmen run between wickets on most hits, potentially scoring multiple runs per ball
- **Bowling restrictions**: Each bowler can only bowl a maximum of 4 overs (24 balls) in T20 format